



ResearchLens Analysis Results - Complete Guide



What is ResearchLens?

ResearchLens is an intelligent research analysis system that automatically reads, understands, and analyzes scientific papers. Think of it like having a super-smart research assistant who can:

1. **Read** dozens of papers and summarize each one
2. **Find patterns** across all papers (topics, themes, methodologies)
3. **Spot gaps** where research hasn't connected different topics yet
4. **Track trends** to see which topics are growing or declining
5. **Create visualizations** to show how research areas relate to each other
6. **Write reports** with AI reasoning about what the gaps mean



How N8N Orchestrates Everything

N8N is an automation platform that coordinates all the analysis steps. Instead of running modules one-by-one on your computer, N8N runs them in parallel across a dedicated workflow:

The N8N Workflow Flow

```
graph TD
    A[User Uploads Papers] --> B[Browser Sends Papers to Backend]
    B --> C[Backend Calls N8N Webhook]
    C --> D[N8N Master Workflow Starts]
    D --> E1[Module 1: Summarize Papers]
    D --> E2[Module 2: Find Topics]
    D --> E3[Module 3: Find Gaps Between Topics]
    D --> E4[Module 4: Track Trends Over Time]
    D --> E5[Module 5: Create Research Map Visualization]
    D --> E6[Module 6: Evidence for Gaps]
    D --> E7[Module 7: Scientific Honesty Score]
    E1 --> F[N8N Returns All Results]
    E2 --> F
    E3 --> F
    E4 --> F
    E5 --> F
    E6 --> F
    E7 --> F
    F --> G[Backend Saves to Database]
    G --> H[Frontend Displays Results]
    H --> I[User Sees Dashboard with 7 Sections]
```

Why use N8N instead of running modules locally?

-  Faster (modules can run in parallel)

-  More reliable (N8N handles errors and retries)
 -  Scalable (can handle bigger datasets)
 -  Separated concerns (frontend doesn't do heavy computation)
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The 10 Analysis Modules Explained

Each module is like a specialist researcher focusing on one specific task:

Module 1: Summarization

What it does: Reads each paper and extracts the most important sentences

Simple analogy: Like reading a book and writing its summary on a post-it note

How it works:

1. Splits each paper into sentences
2. Extracts keywords that appear most frequently
3. Scores each sentence based on how many keywords it contains
4. Keeps the top 3 sentences = summary for that paper

Results stored:

```
{
  paperId: "paper_123",
  title: "Deep Learning in Computer Vision",
  summary: "Three most important sentences from the paper",
  keywords: ["deep learning", "neural networks", "image classification", ...]
}
```

Simple metrics:

- **Count:** How many papers were summarized (usually = number of input papers)
-

Module 2: Topic Modeling

What it does: Groups papers by their main topics/themes

Simple analogy: Like sorting books in a library into different sections (Science, History, Fiction, etc.)

How it works:

1. Takes all paper summaries and keywords
2. Uses machine learning to find groups of similar papers
3. Assigns a semantic label to each group (the topic)
4. Uses Ollama AI to generate human-readable names

Results stored:

```
{
  id: "topic_1",
  name: "Machine Learning and Neural Networks",
  keywords: ["deep learning", "neural networks", "training"],
  coherence: 0.85, // How well papers in this topic "fit together"
  paperIds: ["paper_1", "paper_5", "paper_12"], // Which papers belong here
  llm_label: "AI/ML Foundations", // AI-generated label
  llm_domain_summary: "Core algorithms for neural computation",
  llm_confidence: 0.92
}
```

Metrics you'll see:

- **Coherence** (0-1): How similar the papers in a topic are to each other. Higher = better.
- **Paper Coverage**: How many papers belong to each topic

Module 3: Gap Detection 🔍

What it does: Finds missing connections between different topics

Simple analogy: Like noticing "Wait, why do Topic A (Computer Vision) and Topic B (Natural Language Processing) never talk to each other? They could learn from each other!"

How it works:

1. Takes all topics from Module 2
2. Calculates similarity between each pair of topics
3. Counts how many papers mention both topics together (co-occurrence)
4. Topics that are similar BUT don't co-occur together = a gap
5. Uses Ollama AI to explain what the gap means

Results stored:

```
{
  gapId: "gap_1",
  topicA: "topic_1",
  topicB: "topic_2",
  topicAName: "Machine Learning",
  topicBName: "Knowledge Graphs",
  similarityScore: 0.72, // How related these topics are
  coOccurrence: 0, // How often they appear together
  gapScore: 0.85, // How significant this gap is (0-1)
  explanation: "ML models and knowledge graphs rarely discussed together, despite both being AI paradigms",
  llm_gap_significance: "high",
  llm_integration_opportunity: "Semantic reasoning in neural models"
}
```

Metrics you'll see:

- **Similarity Score** (0-1): How conceptually related two topics are
 - **Co-occurrence**: Number of papers mentioning both topics. 0 = never mentioned together!
 - **Gap Score** (0-1): Importance of this gap. Higher = bigger opportunity for research
-

Module 4: Trend Detection

What it does: Tracks which topics are growing, stable, or declining over time

Simple analogy: Like checking stock market graphs - which research areas are hot right now?

How it works:

1. Groups papers by year
2. Counts how many papers about each topic in each year
3. Detects patterns: Rising (more papers each year), Stable (same), Declining (fewer papers)
4. Uses Ollama AI to explain what the trend means

Results stored:

```
{
  topicId: "topic_1",
  topicName: "Deep Learning",
  trend: "rising", // or "stable" or "declining"
  growthRate: 0.45, // How much is it growing
  peakYear: 2024, // Best year for this topic
  yearlyCounts: [
    { year: 2020, count: 15 },
    { year: 2021, count: 28 },
    { year: 2022, count: 42 },
    { year: 2023, count: 68 }
  ],
  llm_trend_summary: "Deep learning research is accelerating",
  llm_paradigm_shifts: ["From supervised to self-supervised learning"],
  temporalConfidence: 0.88
}
```

Metrics you'll see:

- **Trend**: Rising, Stable, Declining, or Insufficient Data
 - **Growth Rate**: How much the topic is growing
 - **Peak Year**: When this topic had the most papers
-

Module 5: Visualization (Research Map)

What it does: Creates a visual map showing where papers are positioned and how topics relate

Simple analogy: Like a map of a city where each point is a paper, papers in the same topic cluster together, and lines show connections between topics

How it works:

1. Reduces paper data to 2D coordinates using UMAP (a dimensionality reduction algorithm)
2. Plots each paper as a point on the research map
3. Colors points by topic (Machine Learning = blue, NLP = green, etc.)
4. Draws curved lines connecting related topics

Results stored:

```
{
  map: {
    points: [
      {
        paperId: "paper_1",
        title: "Deep Learning Fundamentals",
        x: 150, // Position on canvas
        y: 200,
        topicId: "topic_1",
        topicName: "Deep Learning",
        year: 2023
      },
      // ... more points
    ],
    links: [
      {
        topicAId: "topic_1",
        topicBId: "topic_2",
        strength: 0.65 // How related they are
      },
      // ... more links
    ]
  }
}
```

What you see in the dashboard:

- **Points:** Each is a research paper. Hover to see paper details
- **Colors:** Different colors = different topics
- **Clusters:** Papers bunched together = same topic
- **Curved lines:** Connections between topics (stronger = thicker lines)

Module 6: Gap Evidence

What it does: Finds specific papers that could bridge the gaps

Simple analogy: "We found a gap between Topics A and B. Here are some papers that mention both topics - they could be bridges!"

How it works:

1. Takes each gap from Module 3
2. Searches for papers mentioning both topics
3. Uses Ollama AI to verify they're actually bridging papers
4. Explains how each paper connects the gap

Module 7: Scientific Honesty Score

What it does: Analyzes if papers properly cite related work and avoid plagiarism

Simple analogy: Like a teacher checking if a student properly cited all sources in their essay

How it works:

1. Checks citation patterns
2. Uses Ollama AI to verify claims are properly supported
3. Calculates a score (0-1) indicating research integrity

Results stored:

```
{
  honestyScore: 0.87, // 0-1 scale
  citationCoverage: 0.92, // How many related works are cited
  claimSupport: 0.85, // How well claims are backed by evidence
  analysis: "Generally good research practices with minor citation gaps"
}
```

 The 7 Dashboard Sections (What You See)

After all modules run, the frontend displays results in 7 sections:

Section 1: Dataset Summary

Location: First section in results

What it shows:

- Total number of papers analyzed
- Number of topics found
- Number of gaps identified
- Year range of the papers
- Quality metrics calculated from topic coherence, coverage, and gap novelty

Metrics explained:

Total Papers	Topics	Gaps	Years
--------------	--------	------	-------

145	12	34	20-2024
-----	----	----	---------

- Quality Metrics (0-1 scale):
- Topic Coherence: How well papers in each topic fit together
 - Topic Coverage: What percentage of papers are assigned to topics
 - Gap Novelty: How significant are the gaps found
 - Model Quality: Overall quality score (combination of above)

Why these matter:

- More papers = deeper analysis
- More topics = more diverse research
- More gaps = more opportunities for new research
- High coherence = reliable topic detection

Section 2: Analysis Summary

Location: Second section after dataset summary

What it shows:

- Executive summary of findings (auto-generated by Ollama AI)
- Key findings from the analysis
- Main insights and takeaways

Example summary:

This analysis of 145 papers in AI/ML research (2020-2024) identified 12 major research topics with 34 significant gaps. Key findings include:

1. Deep Learning is rising (68% more papers in 2024 vs 2020)
2. Knowledge Graphs and NLP rarely collaborate (major gap)
3. Most papers follow solid research methodology with strong citations

Section 3: Gap Detection Results

Location: "Main Output" section showing research opportunities

What it shows:

- Ranked list of all gaps found (most important first)
- For each gap:
 - Name: "Topic A × Topic B"
 - Similarity Score: How related the topics are
 - Co-occurrence: How often they appear together
 - Gap Score: How important this gap is

How to read a gap card:

RANK #1: Machine Learning × Knowledge Graphs

Similarity:  0.72 (Very similar topics)

Co-occur:  0 (Never together!)

Gap Score:  0.85 (Major opportunity)

Explanation:

ML models and knowledge graphs are both AI paradigms but rarely discussed together despite potential synergies. This represents a significant research opportunity in semantic reasoning.

Click "View" to see papers that could bridge this gap

Interactive features:

- Click any gap to see details
- View papers that mention both topics
- See AI reasoning about why the gap exists

Section 4: Trend Analysis Results

Location: Shows how topics are growing/declining

What it shows:

- Summary: How many topics are rising, stable, declining
- Charts showing papers per topic over time
- Trend type for each topic

Reading the trends:

TOPIC: Deep Learning

2020 
2021 
2022 
2023 
2024  Trend: RISING ↗
└ Growing at 45% per year

Papers mentioning "deep learning":

2020: 15 papers

2021: 28 papers

2022: 42 papers

2023: 68 papers

What trends mean:

- **Rising** (↗): This area is hot! More researchers working on it
 - **Stable** (→): Established field, consistent interest
 - **Declining** (↘): Less popular now, possibly mature or replaced by newer topics
-

Section 5: Research Map 🗺️

Location: Visual representation of all papers and topics

How to read the map:

Each DOT = one research paper
Dots GROUPED by COLOR = same topic

Machine Learning ●●●●●
(Blue dots close together)

NLP ●●●●
(Red dots close together)

← Papers mentioning both topics are positioned between the groups
Curved LINES between colors = topics are related

Interactive features:

- **Hover over a dot:** See paper title and topic
- **Click a dot:** See full paper details (title, year, main findings)
- **Hover over a line:** See which topics are connected
- **Color legend:** Shows all topics and their colors

What the positions mean:

- **Close dots** = papers very similar to each other
 - **Spaced out dots** = papers somewhat different
 - **Gaps in the map** = research areas not well covered
 - **Lines connecting colors** = topics that reference each other
-

Section 6: Chatbot 💬

Location: Q&A interface about the analysis

What it does:

- Ask questions about the analysis results
- Examples:
 - "What are the top 3 gaps in this research?"
 - "Which topics are rising the fastest?"

- "How good is the research methodology in this corpus?"
- "What papers bridge gap #5?"

How it works:

- Uses Ollama AI (local LLM)
- Answers based on analysis results
- Cites specific papers and gaps

Section 7: Scientific Honesty

Location: Shows research integrity metrics

What it shows:

- Honesty score (0-1)
- Citation coverage: How many related works are cited
- Claim support: How well are claims backed by evidence
- Potential plagiarism detection
- Missing citations warnings

Example:

SCIENTIFIC HONESTY SCORE: 0.87/1.0 ✓

Citation Coverage: 92%
└ How many related works are cited

Claim Support: 85%
└ How well are major claims backed by evidence

Issues Found: 2

- Paper #5 claims not well cited
- Paper #12 possible citation gap in literature review

 Key Metrics Explained

When you see numbers, here's what they mean:

Metric	Range	What it means
Coherence	0-1	How well papers in a topic fit together. 0.8+ is good
Topic Coverage	0-1	What % of papers are grouped into topics. Higher is better
Gap Score	0-1	How important is this gap. 0.7+ is significant
Similarity Score	0-1	How related are two topics. 0.7+ = very related

Metric	Range	What it means
Co-occurrence	0+	How many papers mention both topics. 0 = never together
Growth Rate	%	How much a topic is growing per year
Honesty Score	0-1	Research integrity rating. 0.8+ is good
Confidence	0-1	How confident is the AI in this result

How Results Are Generated & Stored

Step 1: Upload Papers

```
You select PDF files
|
└─> Backend receives papers
    └─> Converts to JSON with title, abstract, content
```

Step 2: Send to N8N

```
Backend sends papers to N8N webhook
|
└─> N8N receives them
    └─> Creates execution record
```

Step 3: Run All Modules

```
N8N runs 7 modules in orchestrated workflow
|
├─> M1 summarizes each paper
├─> M2 groups papers into topics
├─> M3 finds gaps between topics
├─> M4 tracks trends
├─> M5 creates visualization
├─> M6 finds bridge papers
└─> M7 calculates honesty score
```

Step 4: Return Results

```
N8N formats all results
|
└─> Sends back to backend
    └─> Backend saves to MongoDB
```

```
└─> Returns to frontend
    └─> Frontend displays 7 sections
```

Step 5: Save & History

Each analysis is saved with:

- Timestamp
- Paper count
- Topic count
- Gap count
- Year range
- Quality score
- All module results

This is stored in AnalysisRun in MongoDB
and IndexedDB in browser for offline access

Example: Real Analysis Walkthrough

Let's say you upload 100 AI research papers from 2020-2024:

Module 1 Summary: "Found 100 papers. Created summaries for all."

Module 2 Topics: "Found 12 topics:

- Deep Learning (28 papers)
- NLP (22 papers)
- Computer Vision (18 papers)
- Knowledge Graphs (15 papers)
- Reinforcement Learning (12 papers)
- ... (7 more)"

Module 3 Gaps: "Found 34 gaps:

- ML × Knowledge Graphs (opportunity: 0.85)
- NLP × Computer Vision (opportunity: 0.72)
- ... (32 more)"

Module 4 Trend: "Deep Learning: RISING ↗

- 2020: 15 papers
- 2021: 22 papers
- 2022: 28 papers
- 2023: 35 papers Growth rate: 45% per year"

Module 5 Map: "Created visualization with:

- 100 points (papers)

- 12 colored clusters (topics)
- 34 curved lines (gaps between topics)"

Module 7 Honesty: "Average honesty score: 0.87

- Most papers cite relevant work
- Strong methodology in corpus
- 2 papers have citation gaps"

Dashboard Result:

Section 1: 100 papers, 12 topics, 34 gaps, 2020-2024
 Section 2: Summary of key findings
 Section 3: Top 10 gaps with explanations
 Section 4: Trend analysis (Deep Learning rising)
 Section 5: Interactive research map
 Section 6: Chatbot for asking questions
 Section 7: Research integrity metrics
 Section 6: Q&A about findings
 Section 7: Research integrity metrics
 Section 8: Which methods tested on which datasets
 Section 9: Auto-generated related work section

Tips for Interpreting Results

✓ Look For These Patterns:

1. High Gap Scores with 0 Co-occurrence

- Indicates unexplored connection between similar topics
- Best opportunity for new research

2. Rising Trends

- "Hot" research areas
- Good for staying current
- May be competitive

3. High Coherence (0.8+)

- Well-defined topics
- Clear research direction
- Easier to understand

4. Low Similarity with High Co-occurrence

- Disparate topics that work together
- May indicate emerging interdisciplinary field

Watch Out For:

1. Low Coherence (< 0.6)

- Topic may be too broad or poorly defined
- Review papers in that topic manually

2. Very Few Papers in a Topic

- May not be a significant area
- Treat with caution

3. Zero-occurrence Gaps

- Could be genuine opportunities OR
- Topics may be conceptually incompatible

4. Declining Trends

- Topic may be mature (no new problems)
- Or replaced by newer approaches



Using Results for Your Research

Finding Research Questions:

1. Identify high-scoring gaps (Section 3)
2. Find papers that mention both topics (click gap)
3. Ask: "Why haven't these topics been connected?"
4. Propose: "What if we combined them?"

Literature Review:

1. Use the analysis results to find relevant papers
2. Verify accuracy (always check AI output!)
3. Expand with your own insights
4. Cite properly

Identifying Trends:

1. Check Section 4 (Trend Analysis)
2. Find rising topics in your area
3. Review papers from rising topics
4. Position your work within the trend

Spotting Research Integrity:

1. Check Section 7 (Scientific Honesty)
2. Review honesty scores
3. Investigate methodology soundness
4. Ensure proper citations in your own work

Technical Details for Developers

Result Structure (Saved in MongoDB)

```
{
  _id: ObjectId,
  userId: "user_123",
  timestamp: "2024-05-15T10:30:00Z",

  // Summary
  papers: 145,
  topics: 12,
  gaps: 34,
  qualityScore: 0.82,
  yearRange: { start: 2020, end: 2024 },

  // Full results from all modules
  modules: {
    module1: { summaries: [...] },
    module2: { topics: [...], assignments: [...] },
    module3: { gaps: [...] },
    module4: { trends: [...] },
    module5: { map: { points: [...], links: [...] } },
    module6: { gapEvidences: [...] },
    module7: { honestyScore: 0.87, ... }
  },

  // Analysis report
  analysisReport: {
    reportTitle: "Analysis Report",
    reportSummary: "...",
    reportMarkdown: "...",
    executive_summary: "..."
  },

  // Metadata
  status: "completed",
  processingTime: 135000, // milliseconds
  n8nExecutionId: "exec_123"
}
```

Frontend Display Flow

ResultsSection.tsx

- | Section 1: DatasetSummarySection
 - | Section 2: ReportSummarySection
 - | Section 3: GapDetectionSection
 - | Section 4: TrendAnalysisSection
 - | Section 5: ResearchMapResultsSection
 - | Section 6: ChatResultsSection
 - | Section 7: EvaluationSummarySection
 - | Section 8: DatasetMethodMatrixSection
 - | Section 9: RelatedWorkDraftSection
 - | Section 7: EvaluationSummary
1. Extracts data from backendResult.modules
 2. Transforms for display (normalization, filtering)
 3. Renders interactive UI
 4. Handles user interactions (clicks, hovers)

📱 Mobile & Print

Print PDF:

- Click "Print" button in top-right
- Selects content to print
- Generates HTML with all sections
- ****Includes****: Analysis Summary + Quality Metrics + All sections
- Opens print dialog
- Can save as PDF

History:

- All analyses saved in History tab (sidebar)
- Shows timestamp, paper count, topic count, gaps
- Click to view previous analysis
- Delete individual analyses
- Clear all history

✨ Summary

****ResearchLens**** is a complete research analysis system that:

1. ****Reads**** papers (Module 1)
2. ****Understands**** them by finding topics (Module 2)
3. ****Connects**** topics to find gaps (Module 3)
4. ****Tracks**** how topics evolve (Module 4)
5. ****Visualizes**** relationships (Module 5)
6. ****Finds bridges**** for gaps (Module 6)
7. ****Detects issues**** (Modules 7, 8, 10)
8. ****Writes reports**** (Module 9)
9. ****Displays everything**** in 9 interactive sections

****N8N**** orchestrates all modules to run efficiently and reliably.

****The result?**** A comprehensive understanding of any research domain in minutes instead of weeks!

Related Documents

- `N8N\_INTEGRATION\_SUMMARY.md` - How N8N is configured
- `N8N\_WORKFLOW\_QUICK\_REFERENCE.md` - N8N workflow templates
- `N8N\_SETUP\_GUIDE.md` - How to set up N8N
- `README.md` - Project overview
- `WORKFLOWS\_GUIDE.md` - Workflow usage guide

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